

Problem #3

Due: Tuesday, November 11, turn in to canvas, as described in the syllabus. Reminder: We have an in-class midterm on November 18,

1. The following two problems are from your textbook, Chapter 4, Section 4.13 (Exercises):

- (a) Construct a topological conjugacy between the linear systems with the matrices

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

(Hint: Transform to polar coordinates and assume the homeomorphism has the form $h(r, \theta) = (h_r(r), h_\theta(r, \theta))$. The r -dependence of h_θ will involve $\ln(r)$.)

- (b) Use the iterative construction (4.38) of the Hartman-Grobman homeomorphism H to obtain an approximation for the conjugacy for the flow of the system on $\mathbb{R}^3$ given by

$$\begin{aligned}\dot{x} &= -x \\ \dot{y} &= -y + x^2 z \\ \dot{z} &= z\end{aligned}$$

to its linearization at $(0, 0, 0)$. Show that the iteration is not globally convergent. Discuss how to modify the iteration to make it locally convergent, using a “bump function.”

2. Consider the Rössler dynamical system in $\mathbb{R}^3$:

$$\begin{aligned}\dot{x} &= -y - z \\ \dot{y} &= x + ay \\ \dot{z} &= b + z(x - c),\end{aligned}$$

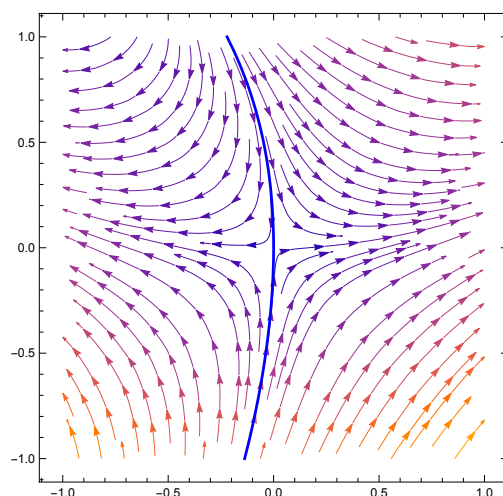
with $a = b = 0.2$.

- (a) Let $c = 2.5$, and find, by numerical solution of the equations, a stable limit cycle. For the next part, you will need to know a point (x_0, y_0, z_0) on the limit cycle to good precision, and also the period T of the limit cycle to good precision. Note those.
- (b) Linearize the equations about the limit cycle and compute the monodromy matrix M , together with its eigenvalues μ , which are the Floquet multipliers. Meiss describes how to do this numerically using the point (x_0, y_0, z_0) as an initial condition for the limit cycle $\gamma(t) = \gamma(T)$, as you found in part (a) (This involves solving a system of six differential equations for three different initial perturbations, which yield the three columns of M .) You should find that one of the Floquet multipliers is 1 to good precision, and the other two satisfy $|\mu| < 1$. Find the eigenvector associated with the eigenvalue $\mu = 1$. Show that it points in a direction that is along the limit cycle at (x_0, y_0, z_0) . (See the discussion about this in the text.)
- (c) Starting with $c = 2.5$, find a stable limit cycle and investigate its stability properties as you gradually increase c . You should find that it loses stability when an eigenvalue of the Monodromy matrix crosses the unit circle at $\mu = -1$. What is the c value associated with this?

- (d) What happens to the limit cycle for c increasing further? (This system exhibits a ‘period-doubling route to chaos’.)
3. This problem relates to the 2-d example at the end of section 5.4, where an approximation to the local stable manifold of the origin is obtained for

$$\begin{aligned}\dot{x} &= 2x + y^2 \\ \dot{y} &= -2y + x^2 + y^2.\end{aligned}$$

- (a) Find an approximation for the local unstable manifold. Plot the phase portrait with both the local stable and unstable manifolds superimposed on it. Meiss shows the stable manifold in his Figure 5.8, and here is my plot generated with Mathematica:



- (b) How might you use your local stable manifold to approximate a global one for this problem? (I can think of a few methods, so this is a bit open-ended.) How well does your approach work?
4. This problem relates to some parts of the proof of Theorem 5.9 (Local Stable Manifold) in the textbook. For full credit on this problem, you need to concisely summarize the relevant parts of the proof in your answer.
- (a) The proof relies on the contraction mapping theorem. Explain its role in the proof. What is the mapping? What is the origin of each of the three terms in $T(x)(t)$ given by Eqn. (5.17)? What is the bound on the contraction rate c obtained in the proof, and how do we ensure $c < 1$?
- (b) Show that any bounded fixed point of the map defined by (5.17) solves the differential equation (5.10). (Hint: this last part is problem 5 on p. 186, where Meiss provides a further hint.)
- (c) Prove the Grönwall inequality that is incorporated in the proof of the local stable manifold theorem, and briefly describe its role in proving Theorem 5.9.